

DISCOVERY PROJECT

PROJECT TITLE

Think of a fun name for your project!

TEAM CONTRACT

Research is better when we share! Read this out loud together and sign your name.

“ We promise to listen to each other. We promise to share the work. If we get stuck, we will ask for help! ”

RESEARCH TEAM

Sign your name here if you promise:

1.

2.

3.

4.

5.

INSPIRATION & TOOLS

An investigator needs a toolkit. Ask an adult to help you scan these codes to get ideas!

Science Ideas

Science Buddies

Look at the pictures to see what other kids have discovered.



Social Studies Ideas

Project Archive

Look at photos of display boards to see how they look.



Bibliography Helper

A **Bibliography** is your "Thank You List." It lists the books and websites that helped you learn. These tools help you write it.

MyBib

A free tool to make your list.



Citation Machine

Ask a parent to help with this!



WORDS TO KNOW

Here are the big words researchers use. Read what they mean in "kid language."

GENERAL TERMS

-  **Abstract:** *"The Short Story."* A tiny summary at the start of your report that tells us what you did.
-  **Bibliography:** *"The Book List."* A list of all the books, websites, and people that taught you the answers.
-  **Discipline:** *"The Subject."* The type of thinking you are using. Are you thinking like a Scientist? Or like a Historian?
-  **Primary Source:** *"The Eyewitness."* Learning from someone who was actually there (like a diary, a letter, or a photo).
-  **Secondary Source:** *"The History Book."* Learning from a book written long after it happened.

SCIENCE WORDS

-  **Hypothesis:** *"My Best Guess."* What I think will happen *before* I do the experiment.
-  **Variable:** *"The Changing Parts."*
 - *Independent Variable:* The **one** thing I change on purpose.
 - *Dependent Variable:* The thing I measure (to see the answer).
 - *Controlled Variable:* The things I keep exactly the same so it is fair.
-  **Engineering Design Process:** How inventors work: Think it, Build it, Test it, Fix it!

SOCIAL STUDIES WORDS

-  **Exposition:** *"Telling a True Story."* A project that explains a person, place, or event.
-  **Anthropology:** *"Studying Culture."* Asking: How do people live and celebrate?
-  **Sociology:** *"Studying Groups."* Asking: How do families and friends act together?
-  **Economics:** *"Studying Money."* Asking: How do we buy, sell, and trade?

PICK YOUR PATH

Your group must choose **ONE** path. What is the most interesting to you?

PATH A: SOCIAL STUDIES

1. Anthropology:

Studying traditions and how people live.

(Ex: Ancient Egypt, Holidays, Native Americans)

2. Economics:

Studying money and jobs.

(Ex: How Money is Made, Trading Cards, Factories)

3. Geography:

Studying maps and the land.

(Ex: Maps of Our City, Volcanoes, National Parks)

4. History:

True stories about the past.

(Ex: Presidents, The First Airplane, Knights)

5. Political Science:

Studying rules and leaders.

(Ex: How Voting Works, The President, Laws)

6. Sociology:

Studying families and communities.

(Ex: Family Traditions, Friendship)

PATH B: SCIENCE

1. Biology:

Studying living things.

(Ex: How Plants Grow, Animal Habits, Germs)

2. Chemistry:

Mixing things to see how they change!

(Ex: Making Slime, Baking, Crystals)

3. Physics:

How things move and fly.

(Ex: Magnets, Gravity, Paper Airplanes)

4. Earth & Environmental:

Studying the planet.

(Ex: Weather, Clean Water, Rocks)

5. Engineering:

Building something to solve a problem.

(Ex: Building a Bridge, Designing a Robot)

✔ **WHAT KIND OF PROJECT?**

Now decide **HOW** you will do your project. Check the box you want to do.

IF YOU CHOSE SOCIAL STUDIES:

- ☐ **The Fixer (Problem Solving)**
Find a problem in your town (like "Litter in the Park") and make a plan to fix it.
- ☐ **The Storyteller (Exposition)**
Tell the true story of a person or event. (Example: "The Story of Video Games").
- ☐ **The Teacher (Demonstration)**
Show us *how* to do something from history (Example: "How to Weave a Basket").

IF YOU CHOSE SCIENCE:

- ☐ **The Tester (Experiment)**
Ask a question, make a guess (hypothesis), and do a fair test to see if you are right.
- ☐ **The Inventor (Engineering)**
Find a problem and build a machine or toy to solve it.

 **Our Topic Title:**

DETECTIVE WORK

Detectives ask questions! We need to answer these small questions to understand our big topic.

Our Main Topic:

5 Keywords (Words we will type into the computer to search):

Question Type	What do we want to know? <i>(Write the question)</i>	Who will find the answer?
WHO? <i>(People)</i>		
WHAT? <i>(Things)</i>		
WHEN? <i>(Dates)</i>		
WHERE? <i>(Places)</i>		
WHY? <i>(Reasons)</i>		
HOW? <i>(How to do it)</i>		

Question Type	What do we want to know? <i>(Write the question)</i>	Who will find the answer?
MYSTERY? <i>(Your Choice!)</i>		

BIBLIOGRAPHY (BOOK LIST)

You must use at least **3 different sources**. Write down the info here so you can give credit!

Source 1: Book

Title of Book:

Author (Who wrote it?):

Company that made the book (Publisher):

Year it was made: _____

Source 2: Website

Title of Article:

Name of Website:

Web Address (URL):

Date You Visited: _____

Source 3: Primary Source

(An interview, a video from the event, a diary, or a visit)

What is it?

Who did you watch or talk to?

Date: _____

Source 4: Book

Title of Book:

Author (Who wrote it?):

Company that made the book:

Year it was made: _____

Source 5: Website

Title of Article:

Name of Website:

Web Address (URL):

Date You Visited: _____

Source 6: Primary Source

What is it?

Who did you watch or talk to?

Date: _____

MY RESEARCH NOTES

Student Name: _____

Don't Copy! Read the sentence, close the book, and then write it in your own words.

What I Learned (<i>Fact or Detail</i>)	Source #	My Thoughts (<i>Why does this matter?</i>)
		I think this is important because...
		This makes me wonder...
		This connects to...
		I was surprised that...

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SCIENCE: THE EXPERIMENT PLAN

Use this page if you are a "Tester" doing an experiment.

1. THE VARIABLES (THE PARTS)

Independent Variable

(The **ONE** thing I will change on purpose):

Dependent Variable

(The thing I will measure or count to see if it worked):

Controlled Variables

(The things I will keep the same so it is a Fair Test):

1. 2.

2. THE HYPOTHESIS (MY PREDICTION)

I predict that if I change the Independent Variable...

THEN the Dependent Variable will...

I think this will happen BECAUSE...

3. MATERIALS LIST

List everything you need. (Example: "2 Cups", "Water", "Ruler")

[illegible]

4. PROCEDURE (STEPS)

Write the steps like a recipe. What do you do 1st, 2nd, 3rd?

[illegible]



SCIENCE: THE INVENTION PLAN

Use this page if you are an "Inventor" building a machine.

1. THE PROBLEM

The problem I want to solve is...

2. CONSTRAINTS (THE RULES)

(Example: "It must be safe," "It must be small," "I can only use glue")

1.

2.

4. TEST & FIX

Test 1: Did it work?

What I will fix for Test 2:

Test 2: Did it work?

What I will fix for Test 3:

Test 3: Did it work?

How I would fix it next time:

SOCIAL STUDIES: THE FIXER

Use this page if you are solving a problem in our town.

1. THE PROBLEM

The problem in our community is...

2. HOW DO I KNOW?

Did you see it? Did you read about it?

I know this is a problem because...

3. THE SOLUTION

My plan to fix this is...

4. WHY IS THIS A GOOD PLAN?

This plan is a good idea because...

SOCIAL STUDIES: THE STORYTELLER

Use this page if you are telling a true story about history or culture.

1. THE TOPIC

I am telling the story of...

This is important for people to know because...

2. THREE PARTS OF THE STORY

Break your story into 3 parts (Beginning, Middle, End).

Part 1: _____

First, I will explain...

Part 2: _____

Next, I will explain...

Part 3: -----

Last, I will explain...

SOCIAL STUDIES: THE TEACHER

Use this page if you are showing us HOW to do something.

1. THE SKILL

I will teach the class how to...

This skill comes from...

2. MATERIALS

What do you need to show us this skill?

3. STEPS

How do you do it?

Step 1: First, you must...

Step 2: Next, you...

Step 3: Finally, you...

4. WHY DO WE DO THIS?

This skill was important because...

SHOWING YOUR DATA

Don't just tell us—SHOW us! Pick a picture to draw so we can see what you learned.

TYPES OF VISUALS



Line Graph: Shows how something changes.
(Example: *How tall a plant grows over 10 days*).



Bar Graph: Compares how much.
(Example: *"How many kids like Pizza vs Tacos"*).



Pie Chart: Shows parts of a whole group.
(Example: *"Half the class likes Blue"*).



Map: Shows places.
(Example: *Drawing a route where explorers traveled*).

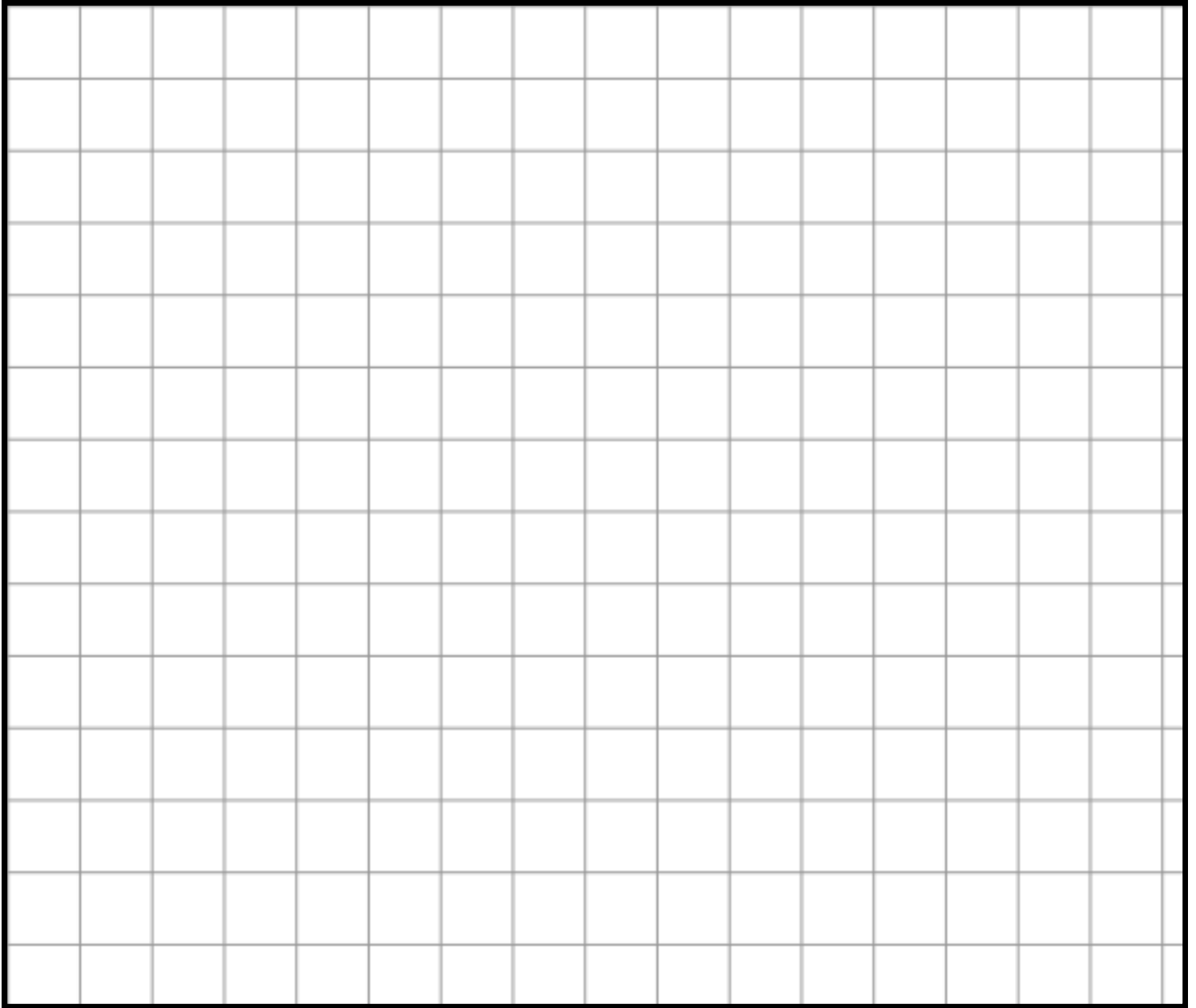


Timeline: Shows what happened first, next, and last.
(Example: *The life of a President*).

DRAW YOUR VISUAL

Use this grid to draw your graph, map, or timeline. Use a ruler to make straight lines!

Title of Visual:



DRAFTING: SCIENCE EXPERIMENT

1. MY QUESTION

My project is about...

I asked the question...

2. WHAT I LEARNED FIRST

I read books and learned that...

3. WHAT I DID

My hypothesis was that IF I change...

THEN this would happen...

To test this, I...

4. MY RESULTS

My data showed that...

5. CONCLUSION

My hypothesis was [right / wrong]...

I think this happened because...

DRAFTING: ENGINEERING

Use this page if you built an invention.

1. THE PROBLEM

I needed to fix a problem where...

2. RESEARCH

I looked at other inventions like...

3. DESIGN & BUILD

My rules (constraints) were...

I built my model using...

4. TESTING

When I tested my first model, it...

I made it better by...

5. CONCLUSION

My final design works because...

DRAFTING: THE FIXER

1. THE TROUBLE

The problem I found is...

2. THE PROOF

I know this is a real problem because...

3. MY IDEA

My plan to fix this is...

4. WHY IT WILL WORK

This is a good idea because...

DRAFTING: THE STORYTELLER

Use this page for Social Studies Stories.

1. MY TOPIC

I am telling the story of...

2. WHAT HAPPENED (THE STORY)

The most important thing to know is...

I also learned that...

3. PRIMARY SOURCE

When I looked at my primary source, I learned...

4. CONCLUSION

This story matters today because...

DRAFTING: THE TEACHER

Use this page for Social Studies Demonstrations.

1. WHAT I WILL SHOW

I am showing the class how to...

2. HISTORY

This skill was used by...

3. THE STEPS

First, you need to...

Then, you must...

4. CONCLUSION

Learning this helps us understand...

THE ABSTRACT (SUMMARY)

This is the short summary everyone reads first. Combine these 3 sentences into one paragraph for your final report!

Part 1: The Question

My project asks the question...

Part 2: What I Did

To find the answer, I...

Part 3: What I Found

I discovered that...

THE FINAL DISPLAY

THE BOARD LAYOUT

Use a tri-fold board. Make sure your letters are **BIG** so we can read them!

LEFT	CENTER	RIGHT
The Plan	The Evidence	The Result
My Question	TITLE	My Data
My Hypothesis	 Photos	Conclusion
Abstract	 Graphs	Next Time...
	My Model	

AM I READY?

Check the boxes when you are done!

☐ **My Topic:** Is it specific? (Not just "Dogs", but "How Dogs Learn").

☐ **My Sources:** Did I use 3 or more sources?

☐ **My Notes:** Did I write facts in my own words? (No Copying!)

☐ **My Report:** Did I use the sentence starters to help me?

☐ **My Visuals:** Did I draw a graph, map, or timeline?

☐ **Spelling:** Did I check my spelling?

THINKING ABOUT MY BRAIN

(Metacognition)

What did you do when you got stuck or frustrated?

What is one new thing you know now that you didn't know before?
